

The memory trail

Lesson 022 — turn a knob, leave a visible record

Each successful software step earns a pulse you can see.

Time	about 45 minutes	Board	Arduino Mega 2560
Prerequisites	Lessons 001–021	Kind	component
Payoffs	live pulse length, eighth-record flourish	Energy	E1, USB only
Physical parts	potentiometer, LED, 1 k Ω resistor	Storage	model only

This lesson is a small record-making machine. Turn a 10 k Ω potentiometer. The Mega samples A0, combines that value with a simulated clock, runs the bytes through recording-only I2C and SPI software, synchronizes a fixed in-memory model, then lights D13. A drier setting makes a short pulse; a wetter setting makes a longer one. Every eighth record earns an unmistakable 1.2-second pulse.

Honest boundary. There is no RTC module, storage card, soil probe, or physical I2C/SPI peripheral in this build. The authorized Elegoo set contains no removable-storage specimen. D20, D21, D49, D50, D51, and D52 remain empty. The sketch generates no bus waveform and proves no physical clock or durable media behavior. A future RTC stage requires the exact module revision, markings, voltage, address, primary datasheet, and electrical review first.

Stop rule. Unplug USB before moving wires. Stop for heat, smell, smoke, repeated USB resets, or any disagreement between the drawing and the part in hand.

Adventure

Build the input first and use its pulse length as evidence. Count eight records to earn the software-restart reward, then keep watching for the fixed safe ending. Each payoff depends on the one before it.

Use the canonical `Lesson022OwnedBuses` sketch throughout. It has a fixed two-minute deadline. At that deadline both indicators go dark, stay inactive for 250 ms, release their resources, and wait for Reset. Turning the knob cannot postpone the ending.

Stage 1: wire one literal circuit

Gather a Mega 2560, USB cable, breadboard, 10 k Ω linear potentiometer, ordinary LED, 1 k Ω resistor, and jumpers. The built-in D13 LED is the activity light; do not build another D13 branch.

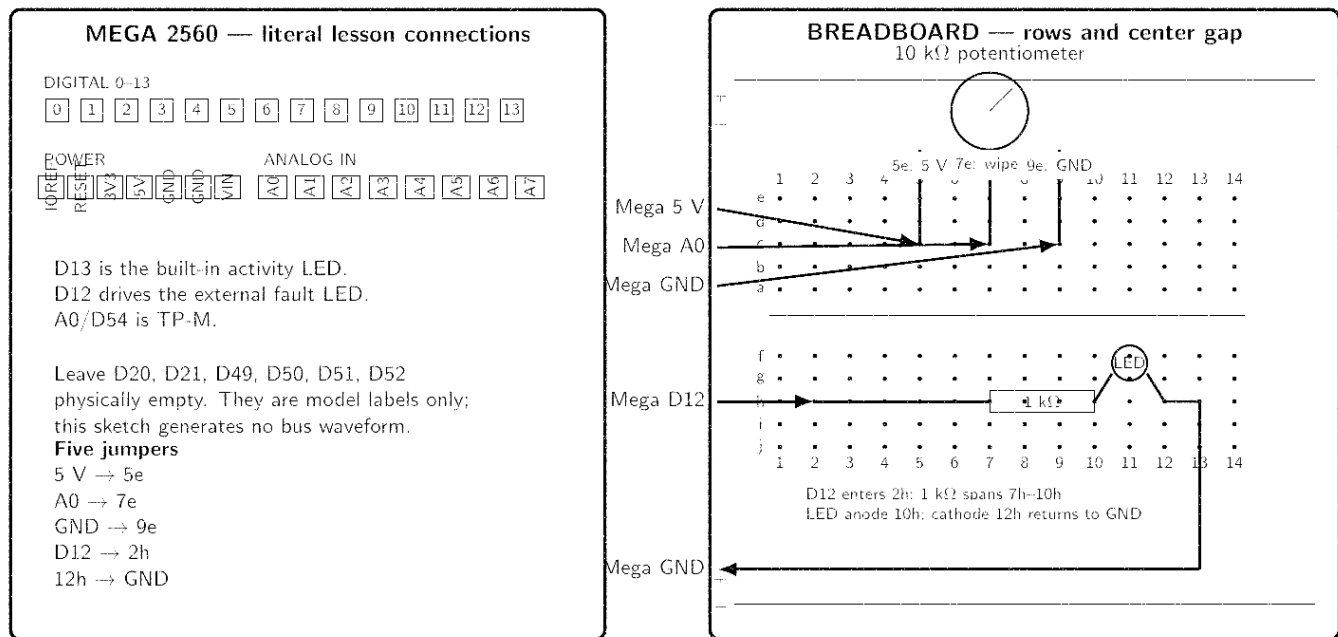


Plate 1: actual headers and breadboard paths. The potentiometer's two outer legs go to 5 V and GND; its center wiper goes to A0. D12 goes through the 1 k Ω resistor to the external LED anode; the cathode returns to the same GND. Verify the potentiometer's center wiper from its own drawing or with a meter before power.

Name	Mega pin	Exact path
TP-M	A0 / D54	potentiometer center wiper
Pot rails	5 V, GND	one outer leg to each rail
Activity	D13	built-in LED only
Fault	D12	D12 → 1 k Ω → LED anode; cathode → GND
Model labels	D20/D21, D49–D52	leave every pin physically empty

Restore USB and upload. After the first 1.5 seconds, D13 should pulse. D12 stays dark. If D12 lights, unplug and inspect only the four physical paths in Plate 1. Adding an RTC or card cannot repair this circuit and is not allowed.

Stage 2: play the pulse instrument

The lesson calibration uses raw ADC counts 200 and 800 as its two chosen endpoints. Counts between them map to 0–1000 permille. This number is a repeatable teaching scale, not soil moisture.

Approximate raw count	Model value	D13 reward
200	0 permille	about 0.1 s
500	500 permille	about 0.5 s
800	1000 permille	about 0.9 s

Turn the shaft slowly from one side to the other. The changing pulse length is the feedback: A0 must be connected, the sample must be valid, the simulated clock state must be valid, both recording transactions must complete, and `sync()` must succeed before D13 turns on.

If the pulse never changes, unplug and check the center wiper at A0 and both outer legs. If it changes backward from your preferred story, swap only the two outer potentiometer legs. Never move the center wiper.

Stage 3: follow one record

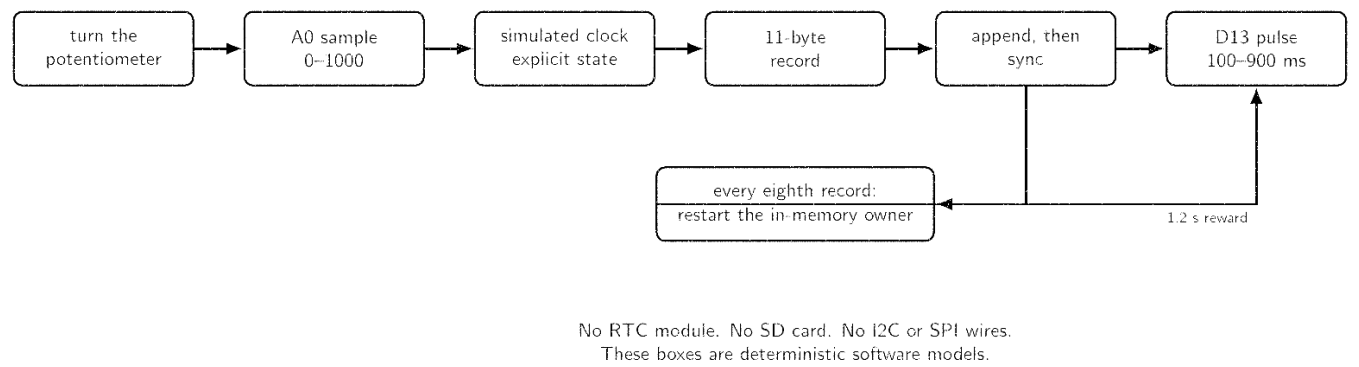


Plate 2: the software journey. It deliberately depicts no RTC board, SD card, or bus wires because none exists in this lesson.

The simulated clock begins at a fixed Unix-seconds value. The 11-byte record contains four clock bytes, one explicit clock-state byte, two raw ADC bytes, two lesson-scale bytes, one sample-state byte, and one synthetic metadata byte. `append()` stages the bytes. Only `sync()` advances the model’s committed prefix. Count the ordinary pulses. The eighth is replaced by a 1.2-second flourish. Before that flourish, the sketch shuts down and reinitializes its `FixedStorage` owner and confirms that its committed byte count is unchanged. That is a software-session restart inside one running sketch. It does not survive Reset or unplugging and must never be described as SD-card or power-loss evidence.

Stage 4: discover the ending

Keep turning the knob if you like. Exactly two minutes after startup, D13 and D12 become inactive. They remain inactive for a quarter second; then the lesson releases its resources and waits. Press Reset to start a new run.

This ending is independent of input. A stuck potentiometer, rapid turning, or the eighth-record reward cannot extend it. The visible result therefore depends on the same lifetime policy exercised by the automated tests behind the scenes.

What the lights actually mean

What you see	What it supports
changing D13 pulses	A0 sampling plus the complete deterministic model path
1.2-second eighth pulse	in-memory owner restart preserved the committed size
D12 latched on	a software status failed; power down and inspect the base circuit
both dark at two minutes	fixed inactive hold and shutdown began

None of those observations supports an I2C acknowledgement, SPI clock edge, RTC oscillator state, SD-card write, media durability, or power-loss retention. Those claims require exact authorized hardware and measurements that this lesson does not contain.

Under the hood

Construction is inert. `initialize()` claims each owner before use, and failure rolls earlier claims back. Every transfer has a bounded timeout. The caller owns each buffer for the whole synchronous call. Shutdown is idempotent and releases owners in reverse order.

```
if (observeControl (now, sample) &&
    persistRecord (now, sample))
{
    acknowledgeRecord (now, sample);
}
```

The reused library value types retain their historical `MoistureSample` names, but this lesson uses them only as a bounded analog calibration model. Its learner-facing input, scale, and record vocabulary is deliberately neutral; no soil or moisture observation occurs.

The host suite retains the exhaustive work: invalid addresses and lengths, claim conflicts, injected configure and transfer failures, chip-select restoration, partial append at every byte, failed sync, exact restart recovery, capacity exhaustion, every clock state, repeated shutdown, destruction, and deterministic replay. Those checks protect the implementation without interrupting the visible adventure.

Make it yours

Try one change at a time: make the ordinary interval one second, change the eighth-record flourish to a two-pulse rhythm, reverse the chosen dry/wet endpoints in software, or draw a paper “memory trail” beside the LEDs. Keep all changes inside this USB-powered, resistor-limited circuit.

Companion material. The HTML page carries searchable pin and model tables plus source links. This printable lesson carries the literal wiring plates, staged visible adventure, and honest model-only hardware boundary.